

Ross Maciejewski

Curriculum Vitae

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EDUCATION

- Ph.D.** *Computer Engineering, Purdue University*, West Lafayette, IN; Dec. 2009; David S. Ebert (PhD advisor); Dissertation: “Exploring Multivariate Data Through the Application of Visual Analytics”
- M.S.** *Computer Engineering, Purdue University*, West Lafayette, IN; Aug. 2004; David S. Ebert (MS advisor); Thesis: “Exploring the Value of Adding Haptic and Stereoscopic Rendering to Volume Modeling”
- B.S.** *Computer Science, University of Missouri*, Columbia, MO; Dec. 2001
- B.S.** *Computer Engineering, University of Missouri*, Columbia, MO; Dec. 2001
- B.S.** *Electrical Engineering, University of Missouri*, Columbia, MO; Dec. 2001

PROFESSIONAL EXPERIENCE

- Jan. 2023 – present** *Director*, School of Computing and Augmented Intelligence, **Arizona State University**, Tempe, AZ
- Aug. 2017 – present** *Director*, Center for Accelerating Operational Efficiency, Department of Homeland Security Center of Excellence
- Aug. 2021 – July 2026** *Ira A. Fulton Professor of Computer Science*, School of Computing and Augmented Intelligence, **Arizona State University**, Tempe, AZ
- Aug. 2021 - present** *Professor*, School of Computing and Augmented Intelligence, **Arizona State University**, Tempe, AZ
- Aug. 2017 – July 2021** *Associate Professor*, School of Computing, Informatics and Decision Systems Engineering, **Arizona State University**, Tempe, AZ
- Aug. 2011 – July 2017** *Assistant Professor*, School of Computing, Informatics and Decision Systems Engineering, **Arizona State University**, Tempe, AZ
- Jan. 2010 – July 2011** *Visiting Assistant Professor*, School of Electrical and Computer Engineering, **Purdue University**, West Lafayette, IN

HONORS & AWARDS

- Oct. 2024** *IEEE VGTC Visualization Academy Inductee*, [Maciejewski, R.](#)
- Mar. 2022** *ACM CHI Honorable Mention*, [Fan, A.](#), [Ma, Y.](#), Mancenido, M., [Maciejewski, R.](#), “Annotating Line Charts in the Wild”
- Feb. 2022** *IEEE Transactions on Visualization and Computer Graphics Best Associate Editor Award, 2021.*
- Aug. 2021** *Ira A. Fulton Professor of Computer Science*, Ira A. Fulton Schools of Engineering, Arizona State University
- Mar. 2018** *Computational Visual Media Journal Honorable Mention*, [Ma, Y.](#), [Chen, W.](#), [Ma, X.](#), [Xu, J.](#), [Huang, X.](#), [Maciejewski, R.](#), [Tung, A.](#), “EasySVM: A Visual Analysis Approach for Open-Box Support Vector Machines”
- Feb. 2018** *ACM CHI Honorable Mention*, [Zhang, J.](#), [Surakitbanharn, C.](#), [Elmqvist, N.](#), [Maciejewski, R.](#), [Qian, Z.](#), [Ebert, D. S.](#), “TopoText: Context-Preserving Text Data Exploration Across Multiple Spatial Scales”
- June 2017** *EuroVis Best Paper*, [Lukasczyk, J.](#), [Maciejewski, R.](#), [Garth, C.](#), [Weber, G.](#), [Leitte, H.](#), “Nested Tracking Graphs”
- May 2017** *Fulton Faculty Exemplar (2017)*, Ira A. Fulton Schools of Engineering, Arizona State University
- May 2017** *Best Researcher Award: Junior Faculty (2016)*, School of Computing, Informatics and Decision Systems Engineering, Arizona State University
- Sept. 2016** *IEEE SciVis Contest 2016 Honorable Mention* for “Viscous Fingers: A Topological Visual Analytics Approach,” awarded at the IEEE VIS Conference ([Lukasczyk, J.](#), [Aldrich, G.](#), [Steptoe, M.](#), [Hamann, B.](#), [Maciejewski, R.](#), [Leitte, H.](#))
- May 2016** *Best Researcher Award: Junior Faculty (2015)*, School of Computing, Informatics and Decision Systems Engineering, Arizona State University
- Oct. 2015** *IEEE-VGTC VPC Visualization Contest Award: Communications Impact* for “ACT Interest-Major Fit” awarded at IEEE VIS ([Wang, F.](#), [Maciejewski, R.](#))

Oct. 2015	<i>VAST Grand Challenge Award: Outstanding Comprehensive Submission</i> for “VADER/VIS-Grand Challenge” awarded at the IEEE Conference on Visual Analytics Science and Technology (Step toe, M. , Krueger, R., Zhang, Y. , Liang, X. , Garcia, R. , Kadambi, S. , Ertl, T., <u>Maciejewski, R.</u>)
May 2015	<i>Best Researcher Award: Junior Faculty (2014)</i> , School of Computing, Informatics and Decision Systems Engineering, Arizona State University
Feb. 2015	<i>IEEE Senior Member</i> , Institute of Electrical and Electronics Engineers
July 2014	<i>NSF Career Award</i> , U.S. National Science Foundation
Oct. 2013	<i>VAST Mini Challenge 1 Award: Excellent Visual Analysis of Structured and Unstructured Data</i> for “Box Office VAST – Team VADER” awarded at the IEEE Conference on Visual Analytics Science and Technology (Lu, Y. , Wang, F. , <u>Maciejewski, R.</u>)
May 2013	<i>U.S. Coast Guard Meritorious Team Commendation</i> as a member of the United States Coast Guard Port Resilience for Operational Tactical Enforcement to Combat Terrorism (PROTECT)
May 2011	<i>Seed for Success</i> , Purdue University, Vice President of Research (\$1M+ grant in AY10-11) awarded for “Development of a Real-Time Infectious Disease Surveillance and Response System for Pakistan”
May 2007	<i>HS-STEM Fellowship</i> , Department of Homeland Security

STUDENT HONORS & AWARDS

Mar., 2017	Rolando Garcia, <i>NSF Graduate Research Fellowship Award</i>
Mar., 2017	Alexandra Porter, <i>NSF Graduate Research Fellowship Award</i>
Dec., 2016	Alexandra Porter, <i>Computing Research Association Undergraduate Research Award – Finalist</i>
Dec., 2016	Rolando Garcia, <i>Computing Research Association Undergraduate Research Award – Honorable Mention</i>
March, 2013	Adam J. Fairfield, <i>Computing Research Association Undergraduate Research Award – Honorable Mention</i>

RESEARCH SUPPORT

Principal Investigator

“Center for Accelerating Operational Efficiency,” *Department of Homeland Security*, \$40M, 8/17 – 8/27, Co-PI (ASU only listed): Pitu Mirchandani, Ronald Askin, Huan Liu, Jingrui He, Hanghang Tong, Jorge Sefair, Giulia Pederelli, Gail-Joon Ahn

“Student Travel Support for the Doctoral Colloquium at IEEE VIS 2018,” *National Science Foundation*, \$25,000, 2/18 – 12/18

“REU/CAREER: Visual analytics algorithms for spatiotemporal analysis,” *National Science Foundation*, \$16,000, 8/17 – 7/18

“INFEWS/T2: Flexible Model Compositions and Visual Representations for Planning and Policy Decisions at the Sub-regional level of the food-energy-water nexus,” *National Science Foundation*, \$2.9M, 8/16 – 8/20, Co-PI: Dave White, Giuseppe Mascaro, Rimjhim Aggarwal, Hessam Sarjoughian

“REU/CAREER: Visual analytics algorithms for spatiotemporal analysis,” *National Science Foundation*, \$16,000, 8/16 – 7/17

“CAREER: Visual analytics algorithms for spatiotemporal analysis,” *National Science Foundation*, \$425,000, 4/14 – 4/19

“A visual analytics framework for the global change assessment model,” *Pacific Northwest National Laboratory*, \$176,420, 1/16-12/18

“WDYTYA: Spatiotemporal network dynamics for community detection,” *Department of Homeland Security VACCINE Center of Excellence*, \$191,707, 8/13-8/16

“Improving predictive analytics capability for VALET,” *Department of Homeland Security VACCINE Center of Excellence*, \$130,186, 12/12 – 8/16

Co-Principal Investigator

“Visualizing High-dimensional Functions in Scientific Machine Learning,” *Department of Energy*, \$312,876 (ASU portion), PI: Gunther Weber (LBNL), Ross Maciejewski (ASU), 9/2022 – 8/2024

“NSF:FAI Towards a Computational Foundation for Fair Network Learning,” *National Science Foundation*, \$585,592, PI: Hanghang Tong (University of Illinois Urbana – Champaign), Co-PI: My Thai (University of Florida), (personal share 30%), 1/2020 – 12/2022.

“Informing Coastal Community Planning and Response to Environmental Change in Regions with Offshore Oil and Gas Operations,” *Gulf Research Program – National Academies*, \$456,000, PI: Tony H. Grubestic, (personal share 50%), 9/15/2016 – 9/14/2018

“Anticipating Effects of Agricultural Trade Networks on Water, Food, and Political Security,” *Skoll Foundation*, \$93,712, PI: Shade Shuttles, (personal share 40%), 3/1/2016 – 2/28/2017

“Foresight Initiative and the new ISR: Interconnectivity, Sustainability, and Resilience,” *National Geospatial Intelligence Agency*, \$4,000,000, PI: Nadya Bliss, (personal share 20%), 6/14 – 10/15

“Development of a Real-Time Infectious Disease Surveillance and Response System for Pakistan: Responding to Human Well-Being in Times of Threat,” *Defense Threat Reduction Agency*, \$1,600,000, PI: Arif Ghafoor, (personal share 5%), 6/10 – 5/13

PUBLICATIONS

Peer-reviewed Journal Articles

Note: Authors who were mentored ASU students at the time of publication are marked in bold.

1. **Hong, J., Seto, C., Fan, A., Maciejewski, R.**, “Do LLMs Have Visualization Literacy? An Evaluation on Modified Visualizations to Test Generalization in Data Interpretation,” *IEEE Transactions on Visualization and Computer Graphics*, (To appear).
2. **Wang, Y., Zhao, J., Hong, J., Askin, R., Maciejewski, R.**, “A Simulation-based Approach for Quantifying the Impact of Interactive Label Correction for Machine Learning,” *IEEE Transactions on Visualization and Computer Graphics*, (To appear).
3. **Xie, T., Chen, J.**, Yang, Y., Geniesse, C., Shi, G., Chaudhari, A., **Cava, J. K.**, Mahoney, M. W., Perciano, T., Weber, G. H., **Maciejewski, R.**, “LossLens: Diagnostics for Machine Learning through Loss Landscape Visual Analytics,” *IEEE Computer Graphics & Applications*, (To appear).
4. Cheng, L., Fu, D., **Maciejewski, R.**, He, J., “DrGNN: Deep Residual Graph Neural Network with Contrastive Learning,” *Transactions on Machine Learning Research*, (To appear).
5. **Hong, J., Hnatyshyn, R.**, Santos, Ebrar A. D., **Maciejewski, R.**, Isenberg, T., “A survey of designs for combined 2D+3D visual representations,” *IEEE Transactions on Visualization and Computer Graphics*, 30(6): 2888-2902, 2024.
6. **Lei, F., Fan, A.**, MacEachren, A. M., **Maciejewski, R.**, “GeoLinter: A Linting Framework for Choropleth Maps,” *IEEE Transactions on Visualization and Computer Graphics*, 30(2): 1592-1607, 2024.
7. **Hong, J., Sheguri, S.**, Askin, R. G., **Maciejewski, R.**, “Carbon emission in football games: Footprint impact of power five conference realignment,” *Sustainability and Sports Science Journal*, 2(2), 98-106, 2024.
8. **Hong, J., Maciejewski, R.**, Trubuil, A., Isenberg, T., “Comparative Visualization of Machine Learning Predictions to Improve Human-AI Teaming on the Example of Cell Lineage,” *IEEE Transactions on Visualization and Computer Graphics*, 30(4): 1956-1969, 2024.
9. **Zhao, J., Wang, Y.**, Mancenido, M. V., Chiou, E. K., **Maciejewski, R.**, “Evaluating the Impact of Uncertainty Visualization on Model Reliance,” *IEEE Transactions on Visualization and Computer Graphics*, 30(7): 4093-4107, 2024.
10. **Lei, F.**, Ma, Y., Fotheringham, A. S., Mack, E. A., Li, Z., Sachdeva, M., Bardin, S., **Maciejewski, R.**, “GeoExplainer: A Visual Analytics Framework for Spatial Modeling Contextualization and Report Generation,” *IEEE Transactions on Visualization and Computer Graphics*, 30(1): 1391-1401, 2024.
11. **Hnatyshyn, R., Zhao, J.**, Ahrens, J., Perez, D., **Maciejewski, R.**, “MolSieve: A Progressive Visual Analytics System for Molecular Dynamics Simulations,” *IEEE Transactions on Visualization and Computer Graphics*, 30(1): 727-737, 2024.
12. Maack, R. G. C., Lukasczyk, J., Tierny, J., Hagen, H., **Maciejewski, R.**, Garth, C., “Parallel Computation of Piecewise Linear Morse-Smale Segmentations,” *IEEE Transactions on Visualization and Computer Graphics*, 30(4): 1942-1955, 2024.
13. Fu, D., Bao, W., **Maciejewski, R.**, Tong, H., He, J., “Privacy-Preserving Graph Machine Learning from Data to Computation: A Survey,” *SIGKDD Explorations*, 25(1): 54-72, 2023.
14. **Xie, T., Ma, Y.**, Kang, J., Tong, H., **Maciejewski, R.**, “FairRankVis: A Visual Analytics Framework for Exploring Algorithmic Fairness in Graph Mining Models,” *IEEE Transactions on Visualization and Computer Graphics*, 28(1): 368-377, 2022.
15. **Zhang, R., Lu, Y.**, Sefair, S., Acevedo, M., Mellin, H., **Maciejewski, R.**, “A Visual Analytics Framework for Conservation Planning Optimization,” *Environmental Modelling and Software*, 145, 2021.
16. Chakrabarti-Bell, S., **Lukasczyk, J.**, Liu, J., **Maciejewski, R.**, Xiao, X., Mayo, S., Regenauer-Lib, K., “Flour Quality Effects on Percolation of Gas Bubbles in Wheat Flour Doughs,” *Innovative Food Science and Emerging Technologies*, 74, 2021.
17. **Zhang, R., Lukasczyk, J.**, Wang, F., Ebert, D., Shakarian, P., Mack, E.A., **Maciejewski, R.**, “Exploring Geographic Hotspots Using Topological Data Analysis,” *Transactions in GIS*, 25(6): 3188-3209, 2021.
18. Mack, E.A., Helderop, E., **Ma, K.**, Grubestic, T.H., Mann, J., Loveridge, S., **Maciejewski, R.**, “A Broadband Integrated Time Series (BITS) for Longitudinal Analyses of the Digital Divide,” *PLoS ONE*, 16(5): e0250732, 2021.
19. **Ma, Y., Medini, P.**, Nelson, J., Wei, R., Grubestic, T., Sefair, J., **Maciejewski, R.**, “A Visual Analytics System for Oil Spill Response and Recovery,” *IEEE Computer Graphics & Applications*, 41(6): 91-100, 2021.
20. **Xie, T., Ma, Y.**, Tong, H., Thai, M., **Maciejewski, R.**, “Auditing the Sensitivity of Graph-based Ranking with Visual Analytics,” *IEEE Transactions on Visualization and Computer Graphics*, 27(2): 1459-1469, 2021.
21. **Ma, Y., Fan, A.**, He, J., Nelakurthi, A., **Maciejewski, R.**, “A Visual Analytics Framework for Explaining and Diagnosing Transfer Learning Processes,” *IEEE Transactions on Visualization and Computer Graphics*, 27(2): 1385-1395, 2021.

22. **Lukasczyk, J.**, Garth, C., **Maciejewski, R.**, Tierny, J., “Localized Topological Simplification of Scalar Data,” *IEEE Transactions on Visualization and Computer Graphics*, 27(2): 572-582, 2021.
23. **Ma, Y.**, **Maciejewski, R.**, “Visual Analysis of Class Separations with Locally Linear Segments,” *IEEE Transactions on Visualization and Computer Graphics*, 27(1): 241-253, 2021.
24. Chen, H., **Soni, U.**, **Lu, Y.**, Huroyan, V., **Maciejewski, R.**, Kobourov, S., “Same Stats, Different Graphs: Exploring the Space of Graphs in Terms of Global Graph Properties,” *IEEE Transactions on Visualization and Computer Graphics*, 27(3): 2056-2072, 2021.
25. **Mathis, B.**, **Ma, Y.**, Mancenido, M., **Maciejewski, R.**, “Exploring the Design Space of Sankey Diagrams for the Food-Energy-Water Nexus,” *IEEE Computer Graphics & Applications*, 41(2): 25-34, 2021.
26. Lee, C., Kim, Y., Jin, S., Kim, D., **Maciejewski, R.**, Ebert, D., Ko, S., “An Interactive Visual Analytics System for Exploring, Monitoring, and Forecasting Road Traffic Congestion,” *IEEE Transactions on Visualization and Computer Graphics*, 26(11): 3133-3146, 2020.
27. Huang, Z., **Lu, Y.**, Mack, E. A., Chen, W., **Maciejewski, R.**, “Exploring the Sensitivity of Choropleths Under Attribute Uncertainty,” *IEEE Transactions on Visualization and Computer Graphics*, 26(8): 2576-2590, 2020.
28. Opejin, A. K., Aggarwal, R. M., White, D. D., Jones, J. L., **Maciejewski, R.**, Mascaro, G., Sarjoughian, H. S., “A Bibliometric Analysis of Food-Energy-Water Nexus Literature,” *Sustainability*, 2020.
29. **Ma, Y.**, **Xie, T.**, Li, J., **Maciejewski, R.**, “Explaining Vulnerabilities to Adversarial Machine Learning through Visual Analytics,” *IEEE Transactions on Visualization and Computer Graphics*, 26(1): 1075-1085, 2020.
30. Lukasczyk, J., Garth, C., Weber, G., Biedert, T., **Maciejewski, R.**, Leitte, H., “Dynamic Nested Tracking Graphs,” *IEEE Transactions on Visualization and Computer Graphics*, 26(1): 249 – 258, 2020.
31. Guan, X., Mascaro, G., Sampson, D., **Maciejewski, R.**, “A Metropolitan Scale Water Management Analysis of the Food-Energy-Water Nexus,” *Science of the Total Environment*, 701(20), 2020.
32. Zhang, S., Tong, H., Xu, J., **Maciejewski, R.**, “Graph Convolutional Networks: A Comprehensive Review,” *Computational Social Networks*, 6(1), 11, 2019.
33. Huang, C.-H., Tong, H., He, J., **Maciejewski, R.**, “Location Prediction for Tweets,” *Frontiers in Big Data*, 2(5), 2019.
34. Gillman, C., Post, T., Wischgoll, T., Hagen, H., **Maciejewski, R.**, “Hierarchical Image Semantics Using Probabilistic Path Propagations for Biomedical Research,” *IEEE Computer Graphics and Applications*, 39(6): 86-101, 2019.
35. Middel, A., Lukasczyk, J., Zakrzewski, S., Arnold, M., **Maciejewski, R.**, “Urban Form and Composition of Street Canyons: A Human-Centric Big Data and Deep Learning Approach,” *Landscape and Urban Planning*, 183:122-132, 2019.
36. **Wang, H.**, **Lu, Y.**, **Steptoe, M.**, Landis, S., Shuttters, S., **Maciejewski, R.**, “A Visual Analytics Framework for Spatiotemporal Trade Network Analysis,” *IEEE Transactions on Visualization and Computer Graphics*, 25(1): 331-341, 2019.
37. Gu, T., Zhu, M., Chen, W., Huang, Z., **Maciejewski, R.**, Chang, L., “Structuring Mobility Transition with an Adaptive Graph Representation,” *IEEE Transactions on Computational Social Systems*, 5(4): 1121-1132, 2018.
38. Middel, A., Lukasczyk, J., Demuz, M., Roth, M., **Maciejewski, R.**, “Sky View Factor Footprints for Urban Climate Modeling,” *Urban Climate*, 25:120-134, 2018.
39. **Soni, U.**, **Lu, Y.**, **Hansen, B.**, Purchase, H., Kobourov, S., **Maciejewski, R.**, “The Perception of Graph Properties in Graph Layouts,” *Computer Graphics Forum*, 37(3), 2018.
40. **Lu, Y.**, **Wang, H.**, Landis, S., **Maciejewski, R.**, “A Visual Analytics Framework for Identifying Topic Drivers in Media,” *IEEE Transactions on Visualization and Computer Graphics*, 24(9): 2501-2515, 2018.
41. Chen, W., Huang, Z., Wu, F., Zhu, M., Guan, H., Zhang, F., **Maciejewski, R.**, “VAUD: A Visual Analysis Approach for Exploring Spatio-Temporal Urban Data,” *IEEE Transactions on Visualization and Computer Graphics*, 24(9): 2535-2648, 2018.
42. **Steptoe, M.**, **Garcia, R.**, Krüger, R., Liang, X., **Maciejewski, R.**, “A Visual Analytics Framework for Exploring Theme Park Dynamics,” *ACM Transactions on Interactive Intelligent Systems*, 8(1), 4, 2018.
43. Kim, S., Jeong, S., Woo, I., Jang, Y., **Maciejewski, R.**, Ebert, D. S., “Data Flow Analysis and Visualization for Spatiotemporal Statistical Data without Trajectory Information,” *IEEE Transactions on Visualization and Computer Graphics*, 24(3): 1287-1300, 2018.
44. White, D. D., Jones, J. L., **Maciejewski, R.**, Aggarwal, R., Mascaro, G., “Stakeholder Analysis for the Food-Energy-Water Nexus in Phoenix, Arizona: Implications for Nexus Governance,” *Sustainability*, 9:2204, 2017.
45. **Wang, F.**, **Hansen, B.**, **Simmons, R.**, **Maciejewski, R.**, “The Name Profiler Toolkit,” *IEEE Computer Graphics & Applications*, 37(5): 61-71, 2017.
46. Luo, W., **Steptoe, M.**, **Zheng, C.**, Link, R., Clarke, L., **Maciejewski, R.**, “The Impact of Spatial Scales on the Inter-Comparison of Climate Scenarios,” *IEEE Computer Graphics & Applications*, 37(5): 40-49, 2017.
47. Andrienko, G., Andrienko, N., Chen, W., **Maciejewski, R.**, Zhao, Y., “Visual Analytics, Movement, and Transportation: State of the Art and Further Research Directions,” *IEEE Transactions on Intelligent Transportation Systems*, 18 (8): 2232 – 2249, 2017.
48. **Lu, Y.**, **Garcia, R.**, **Hansen, B.**, Gleicher, M., **Maciejewski, R.**, “The State-of-the-Art in Predictive Visual Analytics,” *Computer Graphics Forum*, 36(3): 539-562, 2017.

49. Lukaszcyk, J., Maciejewski, R., Garth, C., Weber, G., Leitte, H., “Nested Tracking Graphs,” *Computer Graphics Forum*, 36(3): 12-22, 2017 (Best Paper, EuroVis 2017).
50. Middel, A., Lukaszcyk, J., Maciejewski, R., “Sky View Factors from Synthetic Fisheye Photos for Thermal Comfort Routing – A Case Study in Phoenix, Arizona,” *Urban Planning*, 2(1): 19-30, 2017.
51. Buchanan, V., **Lu, Y.**, McNeese, N., **Steptoe, M.**, Maciejewski, R., Cooke, N., “The Role of Teamwork in the Analysis of Big Data --- A Study of Visual Analytics and Box Office Prediction,” *Big Data*, 5(1): 53-66, 2017.
52. Landis, S., Rezaeedyakenari, S., **Zhang, Y.**, Maciejewski, R., Thies, C., “Fording Differences? Conditions Mitigating Water Insecurity in the Niger River Basin,” *Political Geography*, 56:77-90, 2017
53. **Zhang, Y.**, Maciejewski, R., “Quantifying the Visual Impact of Classification Boundaries in Choropleth Maps,” *IEEE Transactions on Visualization and Computer Graphics*, 23(1): 371-380, 2017.
54. **Wang, F.**, Mack, E.A., Maciejewski, R., “Analyzing Entrepreneurial Social Networks with Big Data,” *Annals of the American Association of Geographers*, 107(1): 130-150, 2017.
55. **Zhang, Y.**, **Luo, W.**, Mack, E.A., Maciejewski, R., “Visualizing the Impact of Geographical Variations on Multivariate Clustering,” *Computer Graphics Forum*, 35(3): 101-110, 2016.
56. **Lu, Y.**, **Steptoe, M.**, **Wang, H.**, Tsai, J., Burke, S., Davulcu, H., Montgomery, D., Corman, S., Maciejewski, R., “Exploring Evolving Media Discourse Through Event Cueing,” *IEEE Transactions on Visualization and Computer Graphics*, 22(1):220-229, 2016.
57. Zhu, B., Zhang, H., Chen, W., Xia, F., Maciejewski, R., “ShotVis: Smartphone-based Visualization of Information from Images,” *ACM Transactions on Multimedia Computing, Communications, and Applications (TOMM)*, 12(1s), 2015.
58. Mack, E. A., Maciejewski, R., “A Profile of Visual Analytical Toolkits for Understanding the Spatio-Temporal Evolution of Broadband Provision,” *Telecommunications Policy Journal*, 39(3-4):320-332, 2015.
59. Malik, A., Maciejewski, R., McCullough, S., Towers, S., Ebert, D., “Proactive Spatiotemporal Resource Allocation and Predictive Visual Analytics for Community Policing and Law Enforcement,” *IEEE Transactions on Visualization and Computer Graphics*, 20(12): 1863-1872, 2014.
60. **Lu, Y.**, **Wang, F.**, Maciejewski, R., “Business Intelligence from Social Media: A Study from the VAST Box-Office Challenge,” *IEEE Computer Graphics and Applications*, 34(5): 58-70, 2014.
61. Malik, A., Maciejewski, R., Jang, Y., Oliveros, S., Yang, Y., Maule, B., White, M., Ebert, D. S., “A Visual Analytics Process for Maritime Response, Resource Allocation and Risk Assessment,” *Information Visualization*, 13(2): 93-110, 2014.
62. Mack, E. A., **Zhang, Y.**, Rey, S., Maciejewski, R., “Spatio-temporal analysis of industrial composition with IVIID: An Interactive Visual Analytics Interface for Industrial Diversity,” *Journal of Geographical Systems*, 16(2): 183-209, 2014.
63. **Zhang, Y.**, Adnan, M., Longley, P., Maciejewski, R., “Exploring Geo-Genealogy Using Internet Surname Search Histories,” *Journal of Maps*, 9(4):481-485, 2013.
64. Kim, S., Maciejewski, R., Malik, A., Jang, Y., Ebert, D. S., Isenberg, T., “Bristle Maps: A Multivariate Abstraction Technique for Geovisualization,” *IEEE Transactions on Visualization and Computer Graphics*, 19(9): 1438-1454, 2013.
65. Maciejewski, R., Woo, I., Jang, Y., Janicke, H., Gaither, K. P., Ebert, D. S., “Abstracting Attribute Space for Transfer Function Exploration and Design,” *IEEE Transactions on Visualization and Computer Graphics*, 19(1): 94-107, 2013.
66. Maciejewski, R., Pattah, A., Ko, S., Hafen, R., Cleveland, W. S., Ebert, D. S., “Automated Box-Cox Transformations for Improved Visual Encoding,” *IEEE Transactions on Visualization and Computer Graphics*, 19(1): 130-140, 2013.
67. Afzal, S., Maciejewski, R., Jang, Y., Elmqvist, N., Ebert, D. S., “Spatial Text Visualization Using Automatic Typographic Maps,” *IEEE Transactions on Visualization and Computer Graphics*, 18(12): 2556-2564, 2012.
68. Woo, I., Maciejewski, R., Gaither, K. P., Ebert, D. S., “Feature-driven Data Exploration for Volumetric Rendering,” *IEEE Transactions on Visualization and Computer Graphics*, 18(10): 1731-1743, 2012.
69. Ko, S., Maciejewski, R., Jang, Y., Ebert, D. “Market Analyzer: An Interactive Visual Analytics System for Analyzing Competitive Advantage Using Point of Sale Data,” *Computer Graphics Forum*, 31(3): 1245-1254, 2012.
70. Livengood, P., Maciejewski, R., Chen, W., Ebert, D. “OmicsVis: An Interactive Tool for Visually Analyzing Metabolomics Data,” *BioMed Central Bioinformatics*, 13(Suppl 8):S6 2012.
71. Maciejewski, R., Livengood, P., Rudolph, S., Collins, T. F., Ebert, D. S., Brigantic, R. T., Corley, C. D., Muller, G. A., Sanders, S. W., “A Pandemic Modeling and Visualization Tool,” *Journal of Visual Languages and Computing*, 22(4):268-278, 2011.
72. Maciejewski, R., Rudolph, S., Hafen, R., Mitchell, M. A., Larew, S. G., Cleveland, W., Ebert, D., “Forecasting Hotspot – A Predictive Analytics Approach,” *IEEE Transactions on Visualization and Computer Graphics*, 17(4): 440-453, 2011.
73. Maciejewski, R., Rudolph, S., Hafen, R., Abusalah, A., Yakout, M., Ouzzani, M., Cleveland, W., Grannis, S., Ebert, D., “A Visual Analytics Approach to Understanding Spatiotemporal Hotspots,” *IEEE Transactions on Visualization and Computer Graphics*, 16(2): 205-220, 2010.

74. Lu, A., Maciejewski, R., Ebert, D., “Volume Composition and Evaluation Using Eye Tracking Data,” *ACM Transactions on Applied Perception*, 7(1): 1-20, January 2010.
75. Maciejewski, R., Chen, W., Woo, I., Ebert, D., “Structuring Feature Space – A Non-Parametric Method for Volumetric Transfer Function Generation,” *IEEE Transactions on Visualization and Computer Graphics*, 15(6): 1473-1480, 2009.
76. Song, Y., Chen, W., Maciejewski, R., Gaither, K., Ebert, D., “Bivariate Transfer Functions on Unstructured Grids,” *Computer Graphics Forum*, 27(3), June 2009.
77. Woo, I., Kim, S., Maciejewski, R., Ebert, D., Ropp, T., Thomas, K., “SDViz: A Context-Preserving Interactive Visualization System for Technical Diagrams,” *Computer Graphics Forum*, 27(3), June 2009.
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Peer-reviewed Conference Articles

Note: Authors who were mentored ASU students at the time of publication are marked in bold.

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2. He, X., Fu, D., Tong, H., Maciejewski, R., He, J., “Temporal Heterogeneous Graph Generation with Privacy, Utility and Efficiency,” *International Conference on Learning Representations (ICLR)*, 2025.
3. Xie, L., Wang, Y., Guan, H., Nag, S., Goel, R., Swamy, N., Yang, Y., Xiao, C., Prisby, J., Maciejewski, R., Zou, J., “IDNet: A Novel Identity Document Dataset via Few-Shot and Quality-Driven Synthetic Data Generation,” *IEEE International Conference on Big Data - Industry and Government Program*, 2024.
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5. **Fan, A., Lei, F.**, Mancenido, M., MachEachren, A., Maciejewski, R., “Understanding Reader Takeaways in Thematic Maps Under Varying Text, Detail, and Spatial Autocorrelation,” *ACM CHI Conference on Human Factors in Computing Systems*, 2024.
6. **Hnatyshyn, R., Hong, J.**, Maciejewski, R., Norby, C., Maley, C. C., “Capturing Cancer as Music: Cancer Mechanisms Expressed through Musification,” *ACM CHI Conference on Human Factors in Computing*, 2024
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9. Kang, T., **Xie, T.**, Wu, X., Maciejewski, R., Tong, H., “InfoFair: Information-Theoretic Intersectional Fairness,” *IEEE Big Data*, 2022.
10. Fu, D., Fang, L., Maciejewski, R., Torvik, V., He, J., “Meta-Learned Metrics over Multi-Evolution Temporal Graphs,” *Knowledge Discovery and Data Mining (KDD)*, 2022.
11. **Fan, A., Ma, Y.**, Mancenido, M., Maciejewski, R., “Annotating Line Charts in the Wild,” *ACM CHI Conference on Human Factors in Computing Systems*, 2022.
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15. Zhou, Y., Nelakurthi, A., Maciejewski, R., Fan, W., He, J., “Crowd Teaching with Noisy Labels,” *The Web Conference (WWW)*, 2020.
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Peer-reviewed Short Papers and Extended Abstracts

1. **Xie, T.**, Geniesse, C., Chen, J., Yang, Y., Morozov, D., Mahoney, M. W., Maciejewski, R., Weber, G. H., "Evaluating Loss Landscapes from a Topology Perspective." *Poster at the NeurIPS 2024 Workshop on Scientific Methods for Understanding Deep Learning (SciForDL)*, 2024.
2. Wu, A., Deng, D., Chen, M., Liu, K., Maciejewski, R., Miksch, S., Strobelt, H., Viegas, F., Wattenberg, M., "Grand Challenges in Visual Analytics Applications," *IEEE Computer Graphics & Applications*, 43(5): 83-90, 2023.
3. Cava, J. K., Shukla, A., Vant, J. W., Karmaker, S. K., Turaga, P. K., Maciejewski, R., Singharoy, A., "Latent Sequence Generation of Steered Molecular Dynamics," *ICLR Workshop on Physics for Machine Learning*, 2023.
4. Ho, N., Cava, J. K., Vant, J., Shukla, A., Miratsky, J., Turaga, P., Maciejewski, R., Singharoy, A., "Learning Free Energy Pathways through Reinforcement Learning of Adaptive Steered Molecular Dynamics," *Machine Learning for Structural Biology (MLSB) Workshop at NeurIPS*, 2022.
5. Fu, D., Ban, Y., Tong, H., Maciejewski, R., He, J., "DISCO: Comprehensive and Explainable Disinformation Detection," *Conference on Information and Knowledge Management (CIKM) Demo Track*, 2022.
6. Colyar, J., Michael, K., Maciejewski, R., Tate, L., "Improving Educational Standards using Visualization Dashboards for Decision Making," *IEEE International Symposium on Technology and Society*, 2022.
7. **Lu, Y., Steptoe, M.**, Buchanan, V., Cooke, N., Maciejewski, R., "Evaluating Forecasting, Knowledge, and Visual Analytics," *The 2nd Workshop on Trust and Expertise in Visual Analytics at IEEE VIS*, 2021.
8. Colyar, J. D., Michael, K., Maciejewski, R., Tate, L., "Constructing a Visualization Dashboard to Improve Educational Standards in Arizona Legislative Districts," *IEEE International Symposium on Technology and Society*, 2021.
9. Maciejewski, R., **Ma, Y., Lukasczyk, J.**, "The Visual Analytics and Data Exploration Research Lab at Arizona State University," *Visual Informatics*, 5(1): 14-22, 2021.
10. Kinner, E., **Lukasczyk, J.**, Rogers, D., Maciejewski, R., Garth, C., "Interpolation of Scientific Image Databases," *International Conference on Physical Modeling for Virtual Manufacturing*, 2020.

11. Gillman, C., Saur, D., Wischgoll, T., Hoffmann, K.-T., Hagen, H., Maciejewski, R., Scheuermann, G., "Uncertainty-aware Brain Lesion Visualization," *Eurographics Workshop on Visual Computing for Biology and Medicine*, 2020.
12. **Lukasczyk, J.**, Garth, C., Larsen, M., Engelke, W., Hotz, I., Rogers, D., Ahrens, J., Maciejewski, R., "Cinema Darkroom: A Deferred Rendering Framework for Large-Scale Datasets," *IEEE Symposium on Large Data Analysis and Visualization*, 2020.
13. Lukasczyk, J., Aldrich, G., **Steptoe, M.**, Hamann, B., Maciejewski, R., Leitte, H., "Viscous Fingers: A Topological Visual Analytics Approach," *IEEE SciVis Contest*, 2016.
14. **Liang, X.**, Cherif, A., **Aggarwal, R.**, Mascaro, G., Gummel, A., Maciejewski, R., "Visualizing Malaria Spread Under Climate Variability," *Proceedings of the Workshop on Visualization in Environmental Science at EuroVis*, 2016.
15. Wu, F., Zhu, M., Zhao, X., Wang, Q., Chen, W., Maciejewski, R., "Visualizing the time-varying crowd mobility," *SIGGRAPH Asia – Higher Performance Computing Workshop*, 2015.
16. Whisner, C., **Porter, A.**, Pecor, N., Maciejewski, R., "A Survey of Personal Nutrition in mHealth Apps," *Personal Visualization: Exploring Data in Everyday Life – IEEE VIS Workshop*, 2015.
17. **Steptoe, M.**, Krueger, R., **Zhang, Y.**, **Liang, X.**, **Luo, W.**, **Garcia, R.**, **Kadambi, S.**, Ertl, T., Maciejewski, R., "VADER/VIS VAST 2015 Grand Challenge Entry," *IEEE Conference on Visual Analytics Science and Technology*, October, 2015.
18. Lukasczyk, J., **Liang, X.**, **Luo, W.**, Ragan, E. D., Middel, A., Bliss, N., White, D., Hagen, H., Maciejewski, R., "A Collaborative Web-Based Environmental Data Visualization and Analysis Framework," *Proceedings of the Workshop on Visualization in Environmental Science at EuroVis*, 2015.
19. Lei, T., **Liang, X.**, Mascaro, G., **Luo, W.**, White, D., Westerhoff, P., Maciejewski, R., "An Interactive Web-Based Geovisual Analytics Tool to Explore Water Scarcity in the Niger River Basin," *Proceedings of the Workshop on Visualization in Environmental Science at EuroVis*, 2015.
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21. **Lu, Y.**, Hu, X., Wang, F., Kumar, S., Liu, H., Maciejewski, R., "Visualizing Social Media Sentiment in Disaster Scenarios," *The 3rd International Workshop on Social Web for Disaster Management (SWDM'15)*, 2015.
22. **Wang, F.**, **Ibarra, J.**, Adnan, M., Longley, P., Maciejewski, R., "What's In a Name? Data Linkage, Demography and Visual Analytics," *Eurovis Workshop on Visual Analytics*, 2014.
23. **Lu, Y.**, **Wang, F.**, Maciejewski, R., "VAST 2013 Mini-Challenge 1: Box Office VAST - Team VADER," *IEEE Conference on Visual Analytics Science and Technology*, October, 2013.
24. Morstatter, F., Kumar, S., Liu, H., Maciejewski, R., "Understanding Twitter Data with TweetXplorer," *Proceedings of the 19th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pp. 1482-1485, 2013.
25. Maciejewski, R., Afzal, S., **Farifield, A. J.**, Ghafoor, A., Ebert, D. S., Ahmed, M., Ayyaz, N., "Enabling Syndromic Surveillance in Pakistan," *In the Abstracts from the 2012 Conference of the International Society of Disease Surveillance*, December 2012.
26. Larew, S. G., Maciejewski, R., Woo, I., Ebert, D., "Spatial Scan Statistics on the GPGPU," *Proceedings of the Visual Analytics in Healthcare Workshop at the IEEE Visualization Conference*, October, 2010.
27. Malik, A., Afzal, S., Hodgess, E., Ebert, D., Maciejewski, R., "VACCINATED – Visual Analytics for Characterizing a Pandemic Spread," *Proceedings of the IEEE Conference on Visual Analytics Science and Technology*, October 2010.
28. Green, T., Maciejewski, R., DiPaola, S., "ALIDA: Using Machine Learning for Intent Discernment in Visual Analytics Interfaces," *Proceedings of IEEE Conference on Visual Analytics Science and Technology*, October 2010.
29. Maciejewski, R., Jang, Y., Ebert, D., Gaither, K., "Abstract Feature Space Representation for Volumetric Transfer Function Exploration," *Dagstuhl Seminar on Scientific Visualization*, July 2010.
30. Maciejewski, R., Brigantic, R., Muller, G., Collins, T. Ebert, D., "Pandemic Visualization Tool for Risk Analysis," *2009 Risk Symposium – Risk Analysis of Complex Systems for National Security Applications*, April 2009.
31. Maciejewski, R., Rudolph, S., Grannis, S., and Ebert, D., "The Day-of-the-Week Effect: A Study Across the Indiana Public Health Emergency Surveillance System," *In the Abstracts from the 2008 Conference of the International Society for Disease Surveillance: Evaluation and Performance*, December 2008.
32. Maciejewski, R., Rudolph, S., Grannis, S., and Ebert, D., "Visual Analytics Toolkit for Evaluating Potential Syndromic Outbreaks," *In the Abstracts from the 2008 Conference of the International Society for Disease Surveillance*, 2008.
33. Maciejewski, R., Rudolph, S., Tebbetts, G., Ebert, D., "Contextualizing Hotspots – A Visual Analytics Approach," *Geospatial Visual Analytics Workshop at the GIScience 2008 Conference*, Park City, Utah, USA, 23-26 September, 2008.
34. Anderson, D., Zheng, C., Maciejewski, R., Hafen, R., Cleveland, W., Ebert, D., Ouzzani, M., Grannis, S., "STL and Local Regression for Modeling Disease Surveillance Counts," *In the Abstracts from the 2007 Conference of the International Society for Disease Surveillance*, December 2007.

35. Pierce, N., Maciejewski, R., Ebert, D., “Visualizing Quantum Dots in a Virtual Environment,” *21st National Conference on Undergraduate Research (NCUR)*, April 2007

Book Chapters

1. Kolak, M., **Steptoe, M.**, Manprizio, H., Azu-Popow, L., Hinchy, M., Malana, G., Maciejewski, R., “Extending Volunteered Geographic Information (VGI) with Geospatial Software as a Service: Participatory Asset Mapping Infrastructures for Urban Health,” *Geospatial Technologies for Urban Health*, *Springer*, 2020.
2. Beigi, G., Hu, X., Maciejewski, R., Liu, H., “An Overview of Sentiment Analysis in Social Media and its Applications in Disaster Relief,” *Sentiment Analysis and Ontology Engineering: An Environment of Computational Intelligence*, *Springer*, 2016.
3. Maciejewski, R., “Geovisualization,” *Handbook of Regional Science*, *Springer*, 2013.
4. Maciejewski, R., “Data Representations, Transformations, and Statistics for Visual Reasoning,” *Morgan & Claypool Publishers*, 2011.

Conference Tutorial Notes

1. Lee, B., Isenberg, P., Stasko, J., Weiskopf, D., Maciejewski, R., “Beyond Paper Types: How to Evaluate and Communicate VIS Research Contributions,” *IEEE VIS*, October, 2019.
2. Ebert, D., Fisher, B., Green, T., Maciejewski, R., “Fundamentals and Applications of Visual Analytics,” *Hawaiian International Conference on System Science*, January 2012.
3. Ebert, D., Fisher, B., Gaither, K., Kielmann, J., Maciejewski, R., “Fundamentals and Applications of Visual Analytics,” *HICSS 2011*, January 2011.
4. Ebert, D., Ertl, T., Keim, D., Koch, S., Maciejewski, R., Thomas, J., Tversky, B., “Fundamentals of Visual Analytics,” *SIGGRAPH 2010*, July 2010.
5. Ebert, D., Ertl, T., Keim, D., Koch, S., Maciejewski, R., Thomas, J., “Applications of Visual Analytics,” *SIGGRAPH 2010*, July 2010.
6. Ebert, D., Maciejewski, R., “From Foundations to Applications of Visual Analytics,” *20th IASTED International Conference on Modeling and Simulation Tutorial Notes*, July 2009.

PATENTS

1. Ebert, D. S., Malik, A., Towers, S., Maciejewski, R., “Proactive spatiotemporal resource allocation and predictive visual analytics system,” U.S. Patent 12073341 issued on Aug. 27, 2024.
2. Frakes, D., Maciejewski, R., Spano, M., Plaas, D., Van Putten, A., Sansone, J., Mortensen, M., Kirkpatrick, N., Thomas, J., “Real-time medical image visualization systems and related methods,” U.S. Patent 10643360 issued on May 5, 2020.
3. Maciejewski, R., Hafen, R., Rudolph, S., Cleveland, W., Ebert, D., “Forecasting hotspots using predictive visual analytics approach,” U. S. Patent 8,924,332 B2 issued on Dec. 30, 2014.
4. Ebert, D. S., Maciejewski, R., Tyner, B., Jang, Y., Cleveland, W., Amass, S., “Animal Symptom Visual Analytics,” U. S. Patent 8882664 B2 issued on November 11, 2014.
5. Ebert, D. S., Collins, T., Maciejewski, R., Malik, A., “Visual Analytics Law Enforcement Tools,” U. S. Patent 8849728 B2 issued on September 30, 2014.

INVITED KEYNOTES

1. Maciejewski, R., “Exploring Spatial Phenomenon with Geovisual Analytics,” *Visualization and Data Analysis*, January, 2021.
2. Maciejewski, R., “Exploring Spatial Phenomenon with Geovisual Analytics,” *ACM SIGSPATIAL*, November, 2020.
3. Maciejewski, R., “Fun with Visualization in the Data Deluge,” *International Symposium on Visual Computing*, October, 2020.
4. Maciejewski, R., “Predictive Visual Analytics: Pitfalls and Potential,” *Pacific VAST*, April, 2019.
5. Maciejewski, R., “Geographic Analysis and Visualization: Fun in the Data Deluge,” *General Motors Geek Week*, September, 2018.
6. Maciejewski, R., “Environmental Visualization under the Lens of Saliency, Credibility and Legitimacy,” *Workshop on Visualization in Environmental Science at EuroVis*, June, 2016.

INVITED SEMINARS

1. Maciejewski, R., “Space, Time and Explainability,” *Scientific Computing and Imaging Institute*, University of Utah, November, 2021.
2. Maciejewski, R., “Visualization, Artificial Intelligence, and Decision Making,” *Data Systems and Foundations Seminar*, Universidad de Los Andes – Bogota, Colombia, November, 2021.
3. Maciejewski, R., “Space, Time and Explainability,” *Geography Colloquium*, University of Colorado – Boulder, March, 2021.
4. Maciejewski, R., “Visualization, Artificial Intelligence, and Decision Making,” *Visual Computing Seminar Series*, Brown University, February, 2021.

5. Maciejewski, R., “Visualization, Artificial Intelligence, and Decision Making,” Data Systems and Foundations Seminar, Universidad de Los Andes – Bogota, Colombia, December, 2020.
6. Maciejewski, R., “Visualization, Artificial Intelligence, and Decision Making,” Data Systems and Foundations Seminar, University of California – Berkeley, October, 2020.
7. Maciejewski, R., “Deceptive Visualizations – Avoiding Pitfalls in Design,” Cancer Prevention and Control Research Network Annual Meeting, February, 2019.
8. Maciejewski, R., “Visual Analytics Methods for Spatiotemporal Analysis,” University of Arizona, September, 2017.
9. Maciejewski, R., “Visual Analytics Methods for Spatiotemporal Analysis,” Institute for Visualization and Interactive Systems (VIS), University of Stuttgart, June 2017.
10. Maciejewski, R., “Integrating a Visual Analytics Curriculum Into Your Own Classroom,” Prairie View A&M University, Texas, June 2017.
11. Maciejewski, R., “Integrating a Visual Analytics Curriculum Into Your Own Classroom,” Florida International University, Florida, May 2016.
12. Maciejewski, R., “Enabling Predictive Analytics Through Visualization,” Los Alamos National Laboratory, March 2016.
13. Maciejewski, R., “Visual Analytics Tools for the Global Change Assessment Model,” Joint Global Change Research Institute, College Park, MD, December, 2015.
14. Maciejewski, R., “Enabling Predictive Analytics Through Visualization,” TU Kaiserslautern, July, 2015.
15. Maciejewski, R., “Integrating a Visual Analytics Curriculum Into Your Own Classroom,” Bethune Cookman University, Florida, June 2015.
16. Maciejewski, R., “Open Source Data Collection, Analysis and Visualization,” *National System for Geospatial Intelligence (NSG) Research & Development Forum (NRF)/GRAND*, April 2015.
17. Maciejewski, R., Olshausen, B. and Kherlopian, A. “NGA Operational Challenges Panel: Immersive Intelligence,” Panel at the NGA Academic Research Program Symposium and Workshops, Washington, D.C., September 2014.
18. Maciejewski, R., “Integrating a Visual Analytics Curriculum Into Your Own Classroom,” Morgan State University, Baltimore, MD, July 2014.
19. Maciejewski, R., “Syndromic Surveillance with Emergency Department Chief Complaints,” Arizona State University School of Public Affairs, March, 2014.
20. Maciejewski, R., “Enabling Predictive Analytics Through Visualization,” University of Arizona, January, 2014
21. Maciejewski, R., “Integrating a Visual Analytics Curriculum Into Your Own Classroom,” Jackson State University, Mississippi, July 2013.
22. Maciejewski, R., “Analytical Brushing for Spatiotemporal Analysis,” University College of London, United Kingdom, July, 2013.
23. Maciejewski, R., “Analytical Brushing for Spatiotemporal Analysis,” City College of London, United Kingdom, July, 2013.
24. Maciejewski, R., “An Introduction to Concepts in Visual Analytics,” *Visual Analytics Summer School at Middlesex University*, United Kingdom, July 2013.
25. Maciejewski, R., “Analytical Brushing for Spatiotemporal Analysis,” Computer Graphics and Visualization Summit, King Abdullah University of Science and Technology (KAUST), Saudi Arabia, April, 2013
26. Maciejewski, R., “Classification Development in NHANES,” Arizona State University School of Nutrition and Health Promotion, January, 2013.
27. Maciejewski, R., “Non-Photorealistic Rendering in Visual Analytics,” Tufts University, September, 2012.
28. Maciejewski, R., “Fundamentals and Applications of Visual Analytics,” Visual Analytics Summer School, Middlesex University, September 2012.
29. Maciejewski, R., “Fundamentals and Applications of Visual Analytics,” Canadian Visual Analytics Summer School, Simon Fraser University, July 2012.
30. Maciejewski, R., “Applications in Geovisual Analytics: Syndromic Surveillance and Crime Mapping,” University College London, Department of Geography, November, 2011.
31. Maciejewski, R., “Applications in Geovisual Analytics: Syndromic Surveillance and Crime Mapping,” Arizona State University School of Geographical Sciences & Urban Planning, September, 2011.
32. Maciejewski, R., “Coordinated Syndromic Surveillance through Visual Analytics,” Pennsylvania State University Center for Infectious Disease Dynamics, April 2011.
33. Maciejewski, R., “Exploring Multivariate Data Through the Application of Visual Analytics,” Arizona State University School of Computing, Informatics and Decision Systems Engineering, April 2011.
34. Ebert, D., Maciejewski, R., “Visual Analytics in Healthcare Applications,” North Carolina Bio-Preparedness Collaborative Webinar, March 2011.
35. Maciejewski, R., “Exploring Multivariate Data Through the Application of Visual Analytics,” Oklahoma State University Department of Computer Science, March 2011.
36. Ebert, D., Maciejewski, R., “From Foundations to Applications of Visual Analytics,” *Visual Analytics Summer School at Middlesex University*, September 2010.
37. Maciejewski, R., “An Introduction to Visual Analytics,” University of Illinois Urbana-Champaign Data Sciences Summer Institute, July 2010.

38. Maciejewski, R., Gates, C., Chaturvedi, A., “The Visualization of Security,” The Center for Education and Research in Information Assurance and Security (CERIAS) Annual Information Security Symposium, March 2010.
39. Ebert, D., Maciejewski, R., “Public Health Data Exploration, Analysis, Modeling and Prediction Using Visual Analytics,” United States Department of Health and Human Services, January 2010.
40. Ebert, D., Maciejewski, R., “Data Modeling and Exploration Using Visual Analytics,” National Defense University, October 2009.
41. Maciejewski, R., “Exploring Multivariate Data Through the Application of Visual Analytics,” Pacific Northwest National Laboratory, April 2011.
42. Maciejewski, R., Görg, C., Tomaszewski, B., “Student Research in Visual Analytics for a Safer Nation,” DHS 2009 University Network Summit, March 2009.
43. Ebert, D., Maciejewski, R., “PURVAC Health Analytics and Visualization Overview,” Center for Disease Control Webinar, August 2008.
44. Amass, S., Maciejewski, R., “Swine Disease, Analysis and Impact,” World Pork Exposition, June 2007.
45. Maciejewski, R., “Linked Animal Human Health Visual Analytics,” IBM's Graphics and Visualization Student Symposium, 2007.

EXTERNAL SERVICE

Organizing Member

2021 – 2024	<i>Co-Chair, Visualization Executive Committee</i>
2020 – 2021	<i>Member, Visualization Executive Committee</i>
2020	<i>Paper Co-Chair, IEEE VAST, IEEE VIS Organizing Committee Member</i>
2019	<i>Paper Co-Chair, IEEE VAST, IEEE VIS Organizing Committee Member</i>
2019	<i>Paper Co-Chair, IEEE PacificVis</i>
2018	<i>Doctoral Colloquium Co-Chair, IEEE VIS Organizing Committee Member</i>
2017	<i>Vice Chair, IEEE VIS</i>
2016	<i>Paper Co-Chair, IEEE Symposium on Large Data Analysis and Visualization</i>
2014, 2016	<i>State-of-the-Art Report Co-Chair, IEEE/VGTC EuroVis Organizing Committee Member</i>
2015	<i>Posters Co-Chair, IEEE/VGTC EuroVis Organizing Committee Member</i>
2015	<i>Workshops Co-Chair, IEEE Conference on Visual Analytics Science and Technology</i>
2012-2013	<i>Posters Co-Chair, IEEE Conference on Visual Analytics Science and Technology</i>

Program Committee Member

2010-2012, 2015-2017	IEEE Conference on Visual Analytics Science and Technology
2012-2014, 2018	IEEE InfoVis
2012-2014, 2017-2019	IEEE/VGTC EuroVis
2009, 2011-2016	Expressive (The joint symposium on Computational Aesthetics, Sketch-Based Interfaces and Modeling, and Non Photorealistic Animation and Rendering)
2014-2015	IEEE Symposium on Large Data Analysis and Visualization
2012-2015	International Conference on Information Visualization Theory and Applications
2017	IEEE/VGTC EuroVAST
2012-2014, 2017	IEEE/VGTC EuroVis Short Papers
2014	IEEE Joint Intelligence and Security Informatics Conference

Workshop Organization

2014	Perer, A., Bertini, E., <u>Maciejewski, R.</u> , Sun, J. (Organizers), “IEEE VIS 2014 Workshop on Visualization for Predictive Analytics”
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Journal Editing

2023 – present	Associate Editor in Chief, <i>IEEE Transactions on Visualization and Computer Graphics</i>
2019 – 2022	Associate Editor, <i>IEEE Transactions on Visualization and Computer Graphics</i>
2017	Guest Associate Editor (Co-Editors B. Hentschel, Hans Hagen, Miriah Meyer), <i>IEEE Computer Graphics & Applications</i> , Special Issue on Applied Visualization
2016	Guest Associate Editor (Co-Editors W. Chen, G. Andrienko, N. Andrienko, Y. Zhao), <i>IEEE Transactions on Intelligent Transportation Systems</i> , Special Issue on Visual Analytics for Intelligent Transportation Systems
2016	Guest Associate Editor (Co-Editors E. Bertini, A. Perer), <i>Big Data</i> , Special Issue on Visualization in Data Science
2015	Guest Associate Editor, <i>Computer Graphics Forum</i> , EuroVis State of the Art Reports

Journal Reviewing (Journals for which I consistently perform reviews)

IEEE Transactions on Visualization and Computer Graphics; IEEE Computer Graphics & Applications

Conference Reviewing (Conferences for which I consistently perform reviews)

IEEE VIS; IEEE/VGTC EuroVis; IEEE PacificVis; ACM CHI

TEACHING EXPERIENCE

Arizona State University (Score provided is “Instructor Score” out of 5.0)

Fall 2023	“SCAI Fulton Fellows Training,” CSE691	44 students	Evaluation Score: 4.42
Fall 2019	“Intro to Undergraduate Research,” FSE 294	16 students	Evaluation Score: 4.58
Fall 2018	“Intro to Undergraduate Research,” FSE 294	12 students	Evaluation Score: 4.86
Fall 2017	“Intro to Undergraduate Research,” FSE 294	6 students	Evaluation Score: 4.78
Spring 2017	“Data Visualization,” CSE 578	117 students	Evaluation Score: 3.98
Fall 2016	“Intro to Data Mining,” CSE 491	38 students	Evaluation Score: 4.67
Fall 2016	“Intro to Undergraduate Research,” FSE 294	9 students	Evaluation Score: 4.76
Fall 2015	“Intro to Undergraduate Research,” FSE 294	29 students	Evaluation Score: 4.86
Fall 2015	“Computer Graphics,” CSE 470/598	28 students	Evaluation Score: 4.70
Fall 2014	“Intro to Undergraduate Research,” FSE 294	10 students	Evaluation Score: 4.57
Fall 2014	“Freshman Engineering,” ASU 101	37 students	Evaluation Score: 4.45
Fall 2014	“Computer Graphics,” CSE 470/598	30 students	Evaluation Score: 4.49
Spring 2014	“Data Visualization,” CSE 591	23 students	Evaluation Score: 4.67
Fall 2013	“Intro to Undergraduate Research,” FSE 294	6 students	Evaluation Score: 4.39
Fall 2013	“Freshman Engineering,” ASU 101	38 students	Evaluation Score: 4.79
Fall 2013	“Computer Graphics,” CSE 470/598	44 students	Evaluation Score: 4.39
Spring 2013	“Data Visualization,” CSE 591	16 students	Evaluation Score: 4.70
Fall 2012	“Freshman Engineering,” ASU 101	38 students	Evaluation Score: 4.62
Fall 2012	“Computer Graphics,” CSE 470/598	44 students	Evaluation Score: 4.52
Spring 2012	“Data Visualization,” CSE 591	24 students	Evaluation Score: 4.46
Fall 2011	“Computer Graphics,” CSE 470/598	20 students	Evaluation Score: 4.30

MENTORING

Postdoctoral Students

Wei Luo (2014-2015); Yafeng Lu (2017 – 2018; First Job at Bloomberg); Yuxin Ma (2018 – 2022); Jonas Lukasczyk (2019 – 2021); Jieqiong Zhao (2020 – 2023); Jiayi Hong (2023 – Present); Tiankai Xie (2023 – present); Rolando Garcia (2024 – present)

PhD Students

Yifan Zhang (2011 - May 2016; Thesis: Visual Analytics for Spatiotemporal Cluster Analysis; First Job at ESRI); Yafeng Lu (2012 – May, 2017; Thesis: Methodologies in Predictive Visual Analytics); Feng Wang (2012 – May 2017; Thesis: Visual Analytics for Networked Based Geographical Phenomena; First Job at GE Research); Hong Wang (2014 – 2019; Thesis: Visual Analytics Methodologies in Causality Analysis; First Job at Uber); Rui Zhang (2017 – 2022; Thesis: Methods for Multiclass Geospatial Data Visualization; First Job at Google); Tiankai Xie (2018 – 2023; Thesis: Explaining the Vulnerabilities of Machine Learning through Visual Analytics); Fan Lei (2019 – 2025; Thesis: Visual Explanation Tools for Spatial Modeling); Arlen Fan (2019 – 2025; Thesis: Addressing Design Errors and Deception in Data Visualizations); Yixuan Wang (2021 – present); Varun Srivastava (2021 – present); Rostyslav Hnatyshyn (2022 – present); Christian Seto (2022 – present); Jiaqing Chen (2022 – present); Jacqueline Nguyen (2024 – present); Shiyi Liu (2024 – present)

Masters Students (Thesis)

Jingxian Ma (2011 – 2012; Thesis: Multimodal Data Fusion as a Predictor of Missing Information in Social Networks; First Job at InfoSol); Thomas Hayden (2013 – 2014; Thesis: Cluster Metrics and Temporal Coherency in Pixel Based Matrices; First Job at Edgenuity); Bi Wu (2011 – 2012; Thesis: Cascading CurtainMap: An Interactive Visualization for Depicting Large and Flexible Hierarchies; First Job at Pay Pal); Zheng Chang (2014 – 2015; Thesis: Visual Analytics Tool for the Global Change Assessment Model; First Job at MathWorks); Michael Steptoe (2014 – 2017; Thesis: Visual Event Cueing in Linked Spatiotemporal Data; Converted to PhD Program); Xing Liang (2014 – 2016; Thesis: Visualizing Numerical Uncertainties in Climate Ensembles; First Job at Amazon); Travis Seville (2016 – 2017; Thesis: A Visual Analytics Process for Exploring Risk and Vulnerability in International Food Trade Networks; First Job at SRP); Utkarsh Soni (2016 – 2018; Thesis: The Perception of Graph Properties in Graph Layouts; Converted to PhD Program); Prannoy Chandra Pydi Medini (2017 – 2018; Thesis: A Spatial Decision Support System for Oil Spill Response and Recover; First Job at Goldman Sachs); Brandon Mathis (2018 – 2019; Thesis: Visualizing Network Structures in the Food, Energy, and Water Nexus; First Job at Oak Ridge National Laboratory); Benjamin Clayton (2018 – 2019; Thesis: The Impact of Graph Layouts on the Perception of Graph Properties); Rostyslav Hnatyshyn (2020 – 2022; Thesis: A Visual Analytics Workflow for Detecting Transition Regions in Large-Scale Molecular Dynamics Simulations)

Thesis Committee Member

PhD: Anjana Arunkumar (Computer Science Ph.D., 2024), Jinbin Huang (Computer Science Ph.D., 2025), Elham Shaabani (Computer Science Ph.D., 2019), Isaac Jones (Computer Science Ph.D., 2019), Arun Reddy (Computer Science Ph.D., 2019), Hu Shao (Geography, Ph.D., 2018), Xuetao Yin (Computer Science Ph.D., 2018), Victor Giroto (Computer Science Ph.D., 2018), Marynia Kolak (Geography, Ph.D., 2018), Saud Alashri (Computer Science Ph.D., 2018), Bardh Hoxha (Computer Science Ph.D., 2017), Shehzad Afzal (Purdue University, 2017), Shamanth Kumar (Computer Science Ph.D., 2015), Fan Bao (Computer Science Ph. D., 2014), Chi-Han Peng (Computer Science Ph. D., 2014), Anshuman Sahu (Industrial Engineering Ph. D., 2013), Ayan Banerjee (Computer Science Ph.D., 2012), Anirudh Kondaveeti (Computer Science Ph.D., 2012), Xun Li (Geography Ph.D.,

2012), Abish Malik (Purdue University, 2014), Insoo Woo (Purdue University, 2012)

Masters: William Boyd (Computer Science MS, 2019), Anique Tahir (Computer Science MS, 2018), Shilpa Nagendar (Computer Science MS, 2017), Verica Buchanan (Human Systems Engineering MS, 2016), Jaeyul Shin (Computer Science MS, 2016), Bolun Li (Computer Science MS, 2014), Savitha Sundaramoorthi (Computer Science MS, 2014), Duyan Ta (Computer Science MS, 2013), Liang Xu (Computer Science MS, 2013), Kunal Khanna (Computer Science MS, 2012), Gustavo Figueroa (Computer Science MS, 2012), Sylvia Oliveros (Purdue University, 2013)

Undergraduate Students

ASU Fulton Undergraduate Research Initiative: Adam Fairfield (2011-2013); Jose Ibarra (2013-2014); Michael Steptoe (2013-2014); Daniel Martin (2014-2015); Alexandra Porter (2014); Sagarika Kadambi (2015-2017); Rolando Garcia (2015-2017); Ryan Kemmer (2019 – 2020); Maya Muir (2021 - 2022); Sia Sheguri (2024 – present); Akshit Jain (2024 – present)

Barrett Honors College Dissertation Advisor: Jose Ibarra (2014); Sagarika Kadambi (2017); Brett Hansen (2017); Maya Muir (2021 - 2022)

Undergraduate Research Assistantship: Christopher Kyle (2015); Sergio Felix (2015); Brett Hansen (2016-2017); Ryan Simmons (2016); Cecilia Nguyen (2017- 2018); Olu Gbatedo (2018 – 2019)

Barrett Honors College Committee Member: Nicholas Ho (2022); Kaan Aksoy (2019); McKinsey Chartier (2015); Allen Hsia (2014); Jordan Nguyen (2014); Anurag Kamasamudram (2012)

DEPARTMENT, SCHOOL AND UNIVERSITY SERVICE

Arizona State University

Fall 2022	<i>Personnel Committee Chair, School of Computing and Augmented Intelligence</i>
Spring 2022	<i>Member, Secure Artificial Intelligence and AI for Social Good Faculty Search - School of Computing and Augmented Intelligence</i>
Fall 2021	<i>Personnel Committee Chair, School of Computing and Augmented Intelligence</i>
Summer 2021	<i>Member, Working Group for SCAI Planning, School of Computing and Augmented Intelligence</i>
Fall 2020	<i>Personnel Committee Chair, School of Computing, Informatics and Decision Systems Engineering</i>
2019 - 2020	<i>Diversity and Inclusion Task Force Initiative, Fulton Schools of Engineering</i>
Spring 2020	<i>Personnel Committee Chair, School of Computing, Informatics and Decision Systems Engineering</i>
2019 – 2022	<i>Personnel Committee Member, School of Computing, Informatics and Decision Systems Engineering</i>
2019 – 2020	<i>Search Committee Member, Data Driven Decision Making Faculty Search – School of Computing, Informatics and Decision Systems Engineering</i>
2018 – 2020	<i>Search Committee Member, Computer Science and Biodesign Institute – Fulton Schools of Engineering</i>
2016 - 2019	<i>Undergraduate Program Committee Chair, School of Computing, Informatics and Decision Systems Engineering</i>
2016 – 2018	<i>Search Committee Chair, Graphics and Visualization Faculty Search – School of Computing, Informatics and Decision Systems Engineering</i>
2015 - 2016	<i>Search Committee Member, Director Search – School of Computing, Informatics and Decision Systems Engineering</i>
2015 - 2016	<i>Search Committee Member, Graphics, Imaging and Vision Search - School of Computing, Informatics and Decision Systems Engineering</i>
2015 - 2016	<i>Undergraduate Program Committee Member, School of Computing, Informatics and Decision Systems Engineering</i>
2012 – 2015	<i>New Faculty Cohort Committee, Fulton Schools of Engineering</i>
2011 - 2015	<i>Graduate Admissions Committee, School of Computing, Informatics and Decision Systems Engineering</i>
2014 - 2015	<i>Search Committee Chair, Big Data Faculty Search - School of Computing, Informatics and Decision Systems Engineering</i>
2012 - 2014	<i>Search Committee Member, Big Data Faculty Search - School of Computing, Informatics and Decision Systems Engineering</i>